CONSTRUCTIBILITY OF R^QF_* OF THE CONSTANT MOD-L SHEAF FOR A PROPER SMOOTH FAMILY OF CURVES WITH GEOMETRICALLY CONNECTED FIBRES

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Allowed standard prerequisites (editorial).

- Proper base change theorem (Etale Cohomology, Theorem etale-cohomology-theorem-proper-base-change, canonical tag 095T). cited at proof lines 4526-4530. Base change compatibility of R^qf_* underwrites the split of S into the l invertible and l zero parts, and its stalk form drives both cases.
- Artin-Schreyer sequence (Etale Cohomology, Section etale-cohomology-section-artin-schreier, canonical tag 0A3J). cited at proof lines 4560-4566. Supplies the sequence 0 -> Z/pZ -> O_X -> O_X -> 0 from which R^1f_* of the constant sheaf is computed in case (b).
- Constructibility of Frobenius kernels and cokernels on vector bundles (Lemma lemma-frobenius-linear-on-vb, canonical tag 0GKU). cited at proof line 4579. The terminal step of case (b): the short exact sequence exhibits R^1f_* as built from Coker(F - 1) and Ker(F - 1) on locally free sheaves, to which the lemma applies.

Source reference. Original lemma 0GKA: sources/stacks-chow/tags.txt, source lines 20680–20680. Cited at proof lines 4567-4570: f_*O_X = O_S and R^1f_*O_X is finite locally free of rank the genus of the fibres of X -> S.

Source reference. Original lemma 0GKC: sources/stacks-chow/tags.txt, source lines 20682–20682. Cited at proof lines 4547-4549: finite sheaves whose specialization maps are isomorphisms are finite locally constant, finishing case (a).

Source reference. Original lemma 0DDF: sources/stacks-chow/tags.txt, source lines 16765–16765. Cited at proof lines 4537-4543: identifies the stalks of R^qf_*Z/lZ with fibre cohomology in cases (a) and (b).

Source reference. Original lemma 0A3H: sources/stacks-chow/tags.txt, source lines 12740–12740. Cited at proof lines 4556-4558: f_*Z/pZ = Z/pZ in case (b) since the fibres are geometrically connected.

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Source reference. Original lemma 0GKD: sources/stacks-chow/tags.txt, source lines 20683–20683. Cited at proof lines 4544-4546: all specialization maps are isomorphisms in case (a).

Source reference. Original reference 0A3J: sources/stacks-chow/tags.txt, source lines 12742–12742. Cited at proof line 4566: source of the Artin-Schreyer sequence 0 -> Z/pZ -> O_X -> O_X -> 0 used in case (b).

Source reference. Original theorem 095T: sources/stacks-chow/tags.txt, source lines 11596–11596. Cited at proof lines 4526-4530: formation of R^qf_*Z/lZ commutes with arbitrary base change, enabling the split into the two cases.

Source reference. Original theorem 03SC: sources/stacks-chow/tags.txt, source lines 4963–4963. Cited at proof lines 4540-4543: finiteness of the stalks of R^qf_*Z/lZ in case (a), and by the same references in case (b).

Source reference. Original lemma 0GKU: sources/stacks-chow/tags.txt, source lines 20699–20699. Cited at proof line 4579: concludes case (b) by applying the lemma to the Coker and Ker of Frobenius-minus-one appearing in the short exact sequence.

Source reference. Original lemma 0EY2: sources/stacks-chow/tags.txt, source lines 18677–18677. Cited at proof lines 4510-4515: after increasing i, f_i has relative dimension 1.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cited at proof lines 4505-4509: descend f to a finitely presented morphism f_i : X_i -> S_i over a finite type Z-subalgebra.

Source reference. Original lemma 0C0C: sources/stacks-chow/tags.txt, source lines 15082–15082. Cited at proof lines 4510-4515: after increasing i, f_i is smooth.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cited at proof lines 4510-4515: after increasing i, f_i is proper.

Source reference. Original lemma 0E0N: sources/stacks-chow/tags.txt, source lines 17543–17543. Cited at proof lines 4516-4518: the open U_i subset S_i over which the fibres of f_i are geometrically connected, used to enter the Noetherian case with the hypothesis in place.

Lemma 0.2. *Let $f : X \to S$ be a proper smooth morphism of schemes with geometrically connected fibres of dimension* 1*. Let ℓ be a prime number. Then $R^q f_* \underline{\mathbf{Z}/\ell\mathbf{Z}}$ is a constructible.*

Proof. We may assume S is affine. Say $S = \mathrm{Spec}(A)$. Then, if we write $A = \bigcup A_i$ as the union of its finite type $\mathbf{Z}$-subalgebras, we can find an i and a morphism $f_i : X_i \to S_i = \mathrm{Spec}(A_i)$ of finite type whose base change to S is $f : X \to S$, see Limits, Lemma 01ZM. After increasing i we may assume $f_i : X_i \to S_i$ is smooth, proper, and of relative dimension 1, see Limits, Lemmas 081F 0C0C, and 0EY2. By More on Morphisms, Lemma 0E0N we obtain an open subscheme $U_i \subset S_i$ such that the fibres of $f_i : X_i \to S_i$ over U_i are geometrically connected. Then $S \to S_i$ maps into U_i. We may replace $X \to S$ by $f_i : f_i^{-1}(U_i) \to U_i$ to reduce to the case discussed in the next paragraph.

Assume S is Noetherian. We may write $S = U \cup Z$ where U is the open subscheme defined by the nonvanishing of ℓ and $Z = V(\ell) \subset S$. Since the formation

of $R^q f_* \underline{\mathbf{Z}/\ell\mathbf{Z}}$ commutes with arbitrary base change (Étale Cohomology, Theorem 095T), it suffices to prove the result over U and over Z. Thus we reduce to the following two cases: (a) ℓ is invertible on S and (b) ℓ is zero on S.

Case (a). We claim that in this case the sheaves $R^q f_* \underline{\mathbf{Z}/\ell\mathbf{Z}}$ are finite locally constant on S. First, by proper base change (in the form of Étale Cohomology, Lemma 0DDF) and by finiteness (Étale Cohomology, Theorem 03SC) we see that the stalks of $R^q f_* \underline{\mathbf{Z}/\ell\mathbf{Z}}$ are finite. By Étale Cohomology, Lemma 0GKD all specialization maps are isomorphisms. We conclude the claim holds by Étale Cohomology, Lemma 0GKC.

Case (b). Here $\ell = p$ is a prime and S is a scheme over $\mathrm{Spec}(\mathbf{F}_p)$. By the same references as above we already know that the stalks of $R^q f_* \underline{\mathbf{Z}/p\mathbf{Z}}$ are finite and zero for $q \geq 2$. It follows from Étale Cohomology, Lemma 0A3H that $f_* \underline{\mathbf{Z}/p\mathbf{Z}} = \underline{\mathbf{Z}/p\mathbf{Z}}$. It remains to prove that $R^1 f_* \underline{\mathbf{Z}/p\mathbf{Z}}$ is constructible. Consider the Artin-Schreyer sequence

$$0 \to \underline{\mathbf{Z}/p\mathbf{Z}} \to \mathcal{O}_X \xrightarrow{F-1} \mathcal{O}_X \to 0$$

See Étale Cohomology, Section 0A3J. Recall that $f_*\mathcal{O}_X = \mathcal{O}_S$ and $R^1 f_* \mathcal{O}_X$ is a finite locally free $\mathcal{O}_S$-module of rank equal to the genera of the fibres of $X \to S$, see Algebraic Curves, Lemma 0GKA. We conclude that we have a short exact sequence

$$0 \to \mathrm{Coker}(F-1 : \mathcal{O}_S \to \mathcal{O}_S) \to R^1 f_* \underline{\mathbf{Z}/p\mathbf{Z}} \to \mathrm{Ker}(F-1 : R^1 f_* \mathcal{O}_X \to R^1 f_* \mathcal{O}_X) \to 0$$

Applying Lemma 0GKU we win. □